

**Topic 8**

# Transformations and vectors

**Concept guide · Worked examples · Practice · Investigation and modelling  
· Solutions**

**Learning focus**

Content explanations, original Cambridge-style practice and an investigation task tailored to International Mathematics 0607.

**Topic map**

|            |                               |
|------------|-------------------------------|
| <b>8.1</b> | Translations and reflections  |
| <b>8.2</b> | Rotations and enlargements    |
| <b>8.3</b> | Combined transformations      |
| <b>8.4</b> | Vectors and magnitude         |
| <b>8.5</b> | Vector geometry and position  |
| <b>8.6</b> | Repeated motions as sequences |

Core foundations and Extended applications are integrated. Practise exact methods, calculator work and the written reasoning required in investigations.

# 1. Moving a shape

A translation moves each point by the same column vector, such as (3,-2): right 3 and down 2. A reflection uses a mirror line and reverses orientation. A rotation requires its centre, angle and direction. Give a transformation in full to distinguish motions with similar images.

## Worked example

Translate A(1,4) by (3,-2):  $A' = (1+3, 4-2) = (4,2)$ .

Exam check: A rotation answer needs the centre as well as its direction and angle.

# 2. Enlargement and scale

An enlargement with scale factor  $k$  sends a point  $X$  from centre  $O$  to  $X'$  with  $OX' = k OX$  as directed. Negative scale factors place images across the centre. Lengths change by  $|k|$  and areas by  $k^2$ . Use lines through corresponding vertices to locate the centre.

## Worked example

Centre origin,  $k = -2$ : (3,1) maps to (-6,-2). An original area of 5 cm<sup>2</sup> becomes 20 cm<sup>2</sup>.

Exam check: A negative scale factor is not a reflection in an arbitrary axis.

# 3. Adding and scaling vectors

Vectors carry magnitude and direction. Add vectors head-to-tail; in component form add matching coordinates.  $A - B$  means  $A + (-B)$ . Multiply a vector by a scalar to change its length and possibly reverse its direction. Magnitude of (a,b) is  $\sqrt{a^2+b^2}$ .

## Worked example

$(2,-3) + (-5,4) = (-3,1)$ ; magnitude of (3,4) is 5.

Exam check: Keep x and y components in their own columns.

# 4. Position vectors and geometry

A position vector runs from the origin to a point. If  $OA = a$  and  $OB = b$ , then  $AB = b - a$ . The midpoint  $M$  of  $AB$  has position vector  $(a+b)/2$ . For collinearity, show vectors are scalar multiples; for parallel lines, show their direction vectors are multiples.

## Worked example

If  $a = (1,2)$  and  $b = (7,8)$ , then  $AB = (6,6)$ , and  $OM = (4,5)$ .

Exam check:  $AB$  and  $BA$  have opposite signs; subtract endpoint minus starting point.

## 5. Combining transformations

Order matters: translate then reflect can yield a different answer from reflect then translate. Apply each operation to the original vertices one at a time, then label the intermediate image. A coordinate grid and matching arrows make the sequence easier to check.

### Worked example

Start with  $(2,1)$ ; reflect in  $y$ -axis to  $(-2,1)$ , then translate by  $(3,-2)$  to  $(1,-1)$ .

Exam check: Do not combine two transformations before checking their order.

## 6. Repeated motions as sequences

Repeating a translation  $v$  from initial position  $p$  gives  $p+n v$  after  $n$  moves. This makes coordinate geometry a simple linear model. Reflection or rotation generally changes direction, so the same additive formula may not work.

### Worked example

From  $(1,1)$  with repeated  $(2,3)$ , after  $n$  moves position is  $(1+2n,1+3n)$ .

Exam check:  $n$  counts complete moves and starts at zero.

## 0607 · Technology and communication

### Graphical calculator and Paper 6

Use coordinates or a grid to verify an image. Record centre, angle, scale factor or translation vector precisely.

State the rule or model, display enough working to reproduce your result, test it against the given values and discuss at least one limitation.



# Practice paper

## Topic 8 · Transformations and vectors

### Instructions

Original practice, informed by June 2026 components. Section A: short multiple choice for revision without a calculator. Section B: show full written working. Section C: investigation and modelling with reasons and checks.

|                 |  |
|-----------------|--|
| Candidate name: |  |
| Class / date:   |  |

**Total: 38 marks**

## Section A · Multiple choice

Choose one answer for each question.

1. Translate (2,3) by (4,-1). [1]

- A (6,2)
- B (-2,4)
- C (8,3)
- D (2,-1)

2. Reflect (3,-2) in the x-axis. [1]

- A (-3,-2)
- B (-3,2)
- C (3,2)
- D (2,-3)

3. Rotate (2,1)  $180^\circ$  about origin. [1]

- A (1,2)
- B (-2,-1)
- C (2,-1)
- D (-1,2)

4. Enlarge (1,-3) by scale factor 2 about origin. [1]

- A (2,-6)
- B (-2,6)
- C (1,-6)
- D (3,-1)

5. Find  $(2,5) + (-3,4)$ . [1]

- A (5,9)
- B (-1,9)
- C (-1,1)
- D (1,20)

6. Magnitude of (6,8)? [1]

- A 7
- B 10
- C 14
- D 100

7. If  $OA = a$  and  $OB = b$ ,  $AB$  equals [1]

- A  $a+b$
- B  $a-b$
- C  $b-a$
- D  $-a-b$

8. For length scale factor -3, area factor is [1]

- A -9
- B -3
- C 3
- D 9

## Section B · Structured questions

9. Triangle  $A(1,1)$ ,  $B(3,1)$ ,  $C(1,4)$  is translated by  $(4,-2)$ .

(a) Give the coordinates of  $A'$  [1]

(b) Give the coordinates of  $B'$  and  $C'$  [2]

(c) Describe what happens to its area. [2]

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**[Total: 5]**

10. Point  $P(2,-1)$  is enlarged about the origin with scale factor  $-3$ .

(a) Find the image  $P'$  [2]

(b) Find  $OP$  and  $OP'$  [2]

(c) State the factor by which area changes. [1]

**[Total: 5]**

11. Position vectors  $OA = (2,3)$  and  $OB = (8,11)$ .

(a) Find  $AB$ . [2]

(b) Find the midpoint  $M$  of  $AB$ . [2]

(c) Find the magnitude of  $AB$ . [2]

**[Total: 6]**

12. Start with point  $X(4,2)$ . Reflect it in the  $y$ -axis, then translate by  $(3,-5)$ .

(a) Find the reflected point  $X'$  [1]

(b) Find the final point  $X''$  [2]

(c) Apply these two transformations in reverse order and find the final point.

[3]

**[Total: 6]**

## Section C · Investigation and modelling

Show the rule, test it with values and explain the result in context.

13. A robot begins at  $(1,1)$  and repeats the translation  $(2,3)$  after each move.

(a) Find its position after one and two moves. [2]

(b) Write its position  $R(n)$  after  $n$  moves. [2]

(c) Find the position after 10 moves. [2]

(d) Explain how an obstacle would limit this model. [2]

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**[Total: 8]**



# Worked solutions

**Fold back this section while completing the practice paper.**

## Marking routine

Method, model, communication and verification matter. A numerical answer should follow from clear working.

## Section A · Answers and reasons

1. A — (6,2). Add the translation vector component by component.
2. C — (3,2). Reflection in the x-axis changes the sign of y alone.
3. B — (-2,-1). A half-turn about the origin negates both coordinates.
4. A — (2,-6). Multiply both coordinates by 2 about the origin.
5. B — (-1,9). Add x-components and y-components separately.
6. B — 10. Magnitude is  $\sqrt{6^2+8^2} = 10$ .
7. C — b-a. Subtract the starting position from the endpoint: b - a.
8. D — 9. Area scales by the square of -3, which is 9.

## Section B · Worked solutions

9. a)  $(1+4, 1-2) = (5, -1)$  [1]. b)  $B' = (7, -1)$ ,  $C' = (5, 2)$  [2]. c) Translation preserves all lengths and area, so the area is unchanged [2].

10. a)  $-3(2, -1) = (-6, 3)$  [2]. b)  $OP = \sqrt{5}$  and  $OP' = \sqrt{45} = 3\sqrt{5}$  [2]. c)  $(-3)^2 = 9$  [1].

11. a)  $b-a = (8-2, 11-3) = (6, 8)$  [2]. b)  $((2+8)/2, (3+11)/2) = (5, 7)$  [2]. c)  $\sqrt{6^2+8^2} = 10$  [2].

12. a)  $(-4, 2)$  [1]. b)  $(-4+3, 2-5) = (-1, -3)$  [2]. c) Translate first:  $(7, -3)$ ; reflect:  $(-7, -3)$  [3].

## Section C · Investigation and modelling solutions

13(a).  $(3, 4)$ ,  $(5, 7)$  [2].

13(b).  $R(n) = (1+2n, 1+3n)$  [2].

13(c).  $(21, 31)$  [2].

13(d). The model assumes every same translation is possible; an obstacle could force a new path [2].

### Self-check

Explain why your model fits the starting data, show an independent check and name a condition under which your prediction would be unreliable.

Reference: Cambridge IGCSE International Mathematics 0607 syllabus 2025-2027, <https://www.cambridgeinternational.org/Images/662472-2025-2027-syllabus.pdf>. Assessment conventions informed by the June 2026 papers and mark schemes supplied by the user. All practice questions are newly written.